

LESSON 5-1 • SESSION 2 • PRACTICE • FORM A**Area of Rhombuses**

Learning target – I can find the area of a rhombus using the base and height or the two diagonals. • 6.G.A.1

REMEMBER

- A **rhombus** has 4 equal sides. It is a parallelogram, so **$A = b \times h$** works.
- With two **diagonals**: **$A = d1 \times d2 \div 2$** .
- Missing diagonal: double the area, then divide ($24 \times 2 = 48$, $48 \div 6 = 8$).

WORKED EXAMPLE

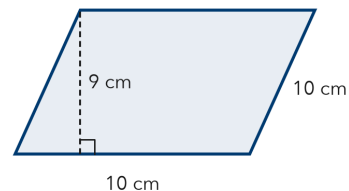
A rhombus has diagonals of 10 cm and 8 cm. Find its area.

Step 1 Use $A = d1 \times d2 \div 2$.**Step 2** $10 \times 8 = 80$ **Step 3** $80 \div 2 = 40$ **Answer** **40 cm²****WORDS** rhombus (*rombo*) • diagonal (*diagonal*) • congruent (*congruente*)**PRACTICE***show your work • label units*

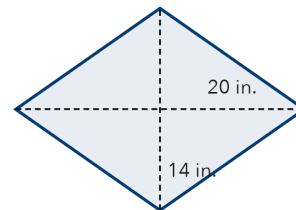
1. Find the area of each rhombus. Choose the formula that fits.

a. diagonals 10 in and 14 in _____**b.** base 9 cm, height 6 cm _____**c.** diagonals 20 ft and 13 ft _____**d.** base 11 cm, height 5 cm _____

- 2.
- Backsplash Tile.**
- A rhombus tile has sides of 10 cm and a height of 9 cm. Find its area.

Hint Any side can be the base.*Show your work.***Answer** _____ cm²

- 3.
- Paper Kite.**
- A kite is a rhombus. Find its area.

Hint Multiply the diagonals, then divide by 2.*Show your work.***Answer** _____ in²

- 4.
- Quilt Patch.**
- A rhombus patch has an area of 84 cm
- ²
- . Its sides are 12 cm. Find its height.

Answer _____ cm

- 5.
- Window.**
- A rhombus window has an area of 150 cm
- ²
- and a height of 10 cm. Find its base.

Answer _____ cm

- 6.
- Sail Panel.**
- A rhombus sail panel has diagonals of 18 ft and 24 ft. Find its area.

*Show your work.***Answer** _____ ft²

7. **Missing Diagonal.** A rhombus sign has an area of 60 m². One diagonal is 12 m. Find the other diagonal.

Hint Double the area, then divide by 12.

Show your work.

Answer_____ m

8. **Bike-Lane Marker.** A rhombus marker has diagonals of 30 in. and 16 in. One strip of tape covers 60 in².

Hint Doubling both diagonals multiplies the area by 2 × 2.

- a. Find the area. _____ in²
- b. How many strips cover it? _____
- c. Both diagonals are doubled. Find the new area. How many times larger is it? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Priya’s Work.** A rhombus has diagonals of 10 cm and 16 cm. Priya wrote: “A = 10 × 16 = 160 cm².” Explain her mistake. Then find the area.

TEST PRACTICE

choose the best answer

10. A rhombus has diagonals of 8 ft and 6 ft. What is its area?

A 14 ft² B 24 ft² C 48 ft² D 96 ft²
11. Which statement is always true about a rhombus?

A All four sides are the same length. B All four angles are right angles.
C Its diagonals are the same length. D It has only one pair of parallel sides.

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

7	15	90	432	216	480	10	140	240	280
O	M	R	Z	B	X	U	H	S	Q
#2		#3	#4	#5	#6	#7	#8a		

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____